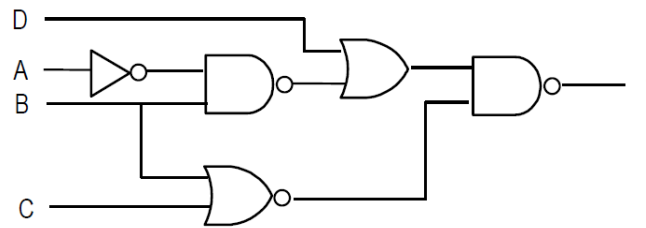


Tutorial 3: Combinational Logic Circuit Fundamental

Question 1

Find the minimum equivalent circuit for following figure using algebraic method.



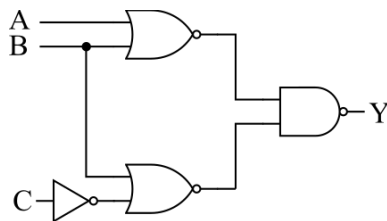
Hint: $f = B + C$

$$\begin{aligned} f(A,B,C,D) &= ((D + (A'B'))(B + C))' \\ &= ((D + A + B')(B'C'))' \\ &= A'BD' + B + C \\ &= B + C \end{aligned}$$

The equivalent circuit is

Question 2

Consider the following logic circuit.



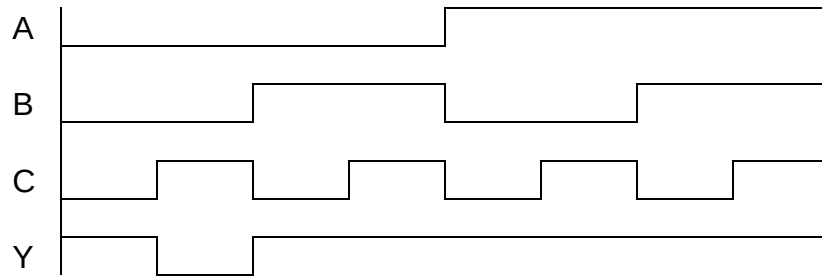
- (i) Derive its Boolean expression.
 - (ii) Determine its truth table.
- Sketch the timing diagram for all possible inputs.

$$\begin{aligned} \text{i) } Y &= ((A+B)' \cdot (B+C'))' \\ &= (A+B) + (B+C') \\ &= A + B + C' \end{aligned}$$

$$\begin{aligned} \text{ii) } Y' &= (A + B + C')' \\ &= A'B'C \end{aligned}$$

A	B	C	Y'	Y
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

iii)



Question 3

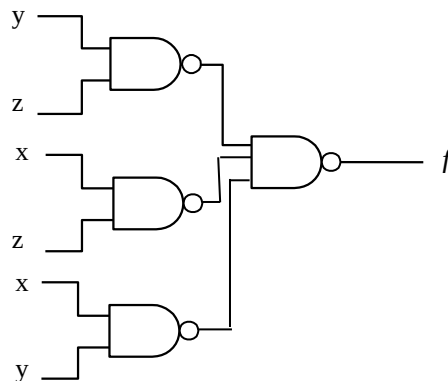
Joe, Jack and Jim get together once a week to either go movie or bowling. To decide, they vote and a simple majority wins. Assuming a vote for the movie is "1", design a minimized logic circuit that automatically computes the decision.

Let $x = \text{Joe}$, $y = \text{Jack}$, $z = \text{Jim}$

Based on simple majority wins $\Rightarrow$ the decision maker set : $\{ 011, 101, 110, 111 \}$

$$\begin{aligned} f(x,y,z) &= x'yz + xy'z + xyz' + xyz \\ &= yz + xz + xy \\ &= [(yz)'(xz)'(xy)']' \end{aligned}$$

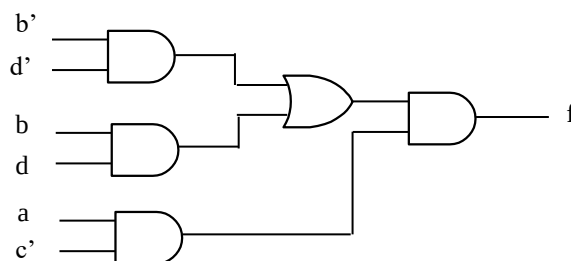
xyz	Y
000	0
001	0
010	0
011	1
100	0
101	1
110	1
111	1



Question 4

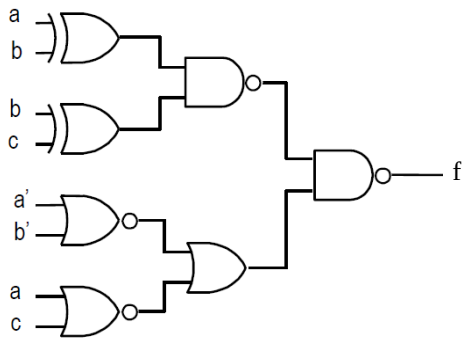
Realize the function $f(a,b,c,d) = \sum m(8, 13)$ using only 2-input AND and OR gates.

$$\begin{aligned} f(a,b,c,d) &= \sum m(8, 13) \\ &= ab'c'd' + abc'd \\ &= ac'(b'd' + bd) \end{aligned}$$



Question 5

Using algebraic method, find simplified expression and 2-level circuit for following.



Hint: $f = a'c + (a \oplus b) = a'c + a'b + ab'$

$f(a,b,c)$

$$\begin{aligned}
 &= \overline{(a \oplus b)(b \oplus c)} \cdot \overline{(a + b + a + c)} \\
 &= (a \oplus b)(b \oplus c) + \overline{(a + b + a + c)} \\
 &= (ab' + a'b)(b'c + bc') + (a' + b')(a + c) \\
 &= ab'c + a'bc' + b'c + a'c + ab' \\
 &= a'bc' + b'c + a'c + ab' \\
 &= a'bc' + a'c + ab' \quad \text{as } b'c + a'c + ab' = a'c + ab' \\
 &= a'(bc' + c) + ab' \quad \text{; expand then combine} \\
 &= a'(b + c) + ab' \\
 &= a'b + a'c + ab' \\
 &= a'c + (a \oplus b)
 \end{aligned}$$

